# PAVEL HASAN JOY

**Aspiring Full-Stack Software Engineer & ML/AI Engineer**

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## PROFESSIONAL SUMMARY

Undergraduate Computer Science & Engineering student at the University of Liberal Arts Bangladesh (ULAB) with a proven track record of engineering impactful software solutions from first principles. Experienced in building 3D geospatial observatories (NASA Space Apps Challenge), high-performance information retrieval engines with sub-3ms latency, and campus-wide networking portals. Proficient in C, C++, Java, Python, and modern web frameworks (React, Next.js). Known for disciplined algorithmic problem-solving, clean architecture, and rapid adaptability in high-pressure hackathon environments.

## EDUCATION

**B.Sc. in Computer Science & Engineering (CSE)** — **2026 – Present**
University of Liberal Arts Bangladesh (ULAB) — Dhaka, Bangladesh
- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (C++/Java), Systems Design, Discrete Mathematics.

**Higher Secondary Certificate (HSC) in Science** — **2023 – 2024**
Naogaon Govt. College — Naogaon, Rajshahi, Bangladesh — **GPA: 4.50 / 5.00**

**Secondary School Certificate (SSC) in Science** — **2021 – 2022**
Tapir Bari Ansar High School — Gazipur, Dhaka, Bangladesh — **GPA: 5.00 / 5.00 (Golden)**

## TECHNICAL SKILLS & COMPETENCIES

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|---|---|
| **Programming Languages:** | C, C++, Java, Python, JavaScript (ES6+), TypeScript |
| **Web & Frameworks:** | React, Next.js, Node.js, FastAPI, Tailwind CSS, HTML5, CSS3 |
| **Core Engineering:** | Data Structures & Algorithms, Object-Oriented Programming (OOP), REST APIs |
| **Developer Tools:** | Git, GitHub, VS Code, Linux/Bash, Vercel, npm |
| **Spoken Languages:** | Bangla (Native), English (Professional Working), Hindi (Conversational) |
| **Soft Skills:** | Problem Solving, Teamwork & Collaboration, Fast Learner, Time Management |

## FEATURED ENGINEERING PROJECTS

**Climate Lens — 3D NASA Space Apps Climate Observatory** — github.com/pavel-hasan-joy/Climate-Lens
*Technologies: TypeScript, Next.js, NASA POWER API, geoBoundaries, Tailwind CSS*
- Built an interactive 3D climate observatory web application for the NASA Space Apps Challenge to visualize and analyze multi-decadal global environmental datasets.
- Designed an intuitive 3D spatial user interface delivering real-time insights into climate volatility, rainfall anomalies, and root-zone soil moisture across all 64 districts in Bangladesh.

**ULAB Setu — University-Wide Networking & Alumni Portal** — github.com/pavel-hasan-joy/ulab-setu
*Technologies: Next.js 16, React 19, TypeScript, Prisma ORM, Tailwind CSS*
- Developed a centralized university networking and community portal designed to connect ULAB students, faculty, and alumni on a unified platform.
- Engineered role-based access control (RBAC), academic collaboration feeds, peer-to-peer resource sharing, and verified alumni referral job board integration.

**Bangla & Banglish Search Engine — Information Retrieval Core** — github.com/pavel-hasan-joy/Bangla-Search-Engine
*Technologies: Python (Standard Library), Algorithms, FastAPI, Information Retrieval, BM25*
- Engineered an ultra-fast phonetic and BM25 search engine from scratch in pure Python standard library with zero external runtime dependencies.
- Implemented 3-tier G2P phonetic hashing, trigram-indexed Damerau-Levenshtein fuzzy matching, and Block-Max WAND dynamic pruning delivering sub-3ms latency over 100,000 documents.

**Campus Safety — Real-Time Emergency Response Platform** — github.com/pavel-hasan-joy
*Technologies: JavaScript, React, WebSockets / Real-time APIs, CSS3*
- Engineered a crisis coordination platform during the ULAB CPC Hackathon to assist students and emergency coordinators during campus emergencies.
- Integrated instant SOS alert routing, student status verification, and responsive emergency service locator.

## CERTIFICATIONS

**Introduction to AI and Machine Learning on Google Cloud** — Google Cloud / NetCom Learning

## HONORS & HACKATHON ACHIEVEMENTS

- **4th Position — ULAB Computer Programming Club Hackathon (15-08-2026)**
  Secured 4th position among university engineering teams by designing and deploying Campus Safety, an emergency web platform to help students during critical incidents.
- **Global Hackathon Participant — NASA Space Apps Challenge (2026)**
  Architected Climate Lens — Bangladesh, a 3D spatio-temporal geospatial observatory evaluating climate vulnerability from NASA Earth observation satellites and CMIP6 climate models.